THE INDIAN INTERNATIONAL SCHOOL,DSO

ırksheet – Chapter 2: Inverse Trigonometric Functions (2026-2027)

Grade: 12 Subject: MATHEMATICS

Name: _____

Date: _____

Instructions:

- This worksheet has 20 questions arranged in three levels of increasing difficulty.
- Level 1: Q 1 – Q 7 (MCQs, Assertion–Reason and short answer questions)
- Level 2: Q 8 – Q 14 (short answer questions)
- Level 3: Q 15 – Q 20 (long answer and case-based questions)
- Marks for each question are given against it. All questions are compulsory.
- The answer key is given at the end of the worksheet.

Q No.	Question	Marks
	LEVEL 1 **Q 1 – Q 7**	**8**
1.	The principal value of $\sin^{-1}\left(-\frac{1}{2}\right)$ is: (A) $-\frac{\pi}{6}$ (B) $\frac{7\pi}{6}$ (C) $\frac{5\pi}{6}$ (D) $\frac{\pi}{6}$	1
2.	The value of $\cos^{-1}\left(\cos\frac{7\pi}{6}\right)$ is: (A) $\frac{7\pi}{6}$ (B) $\frac{5\pi}{6}$ (C) $\frac{\pi}{6}$ (D) $-\frac{\pi}{6}$	1
3.	The domain of the function $y = \sin^{-1}(2x)$ is: (A) $[-1,1]$ (B) $\left[-\frac{1}{2},\frac{1}{2}\right]$ (C) $[-2,2]$ (D) $[0,1]$	1
4.	The range of the principal value branch of $\sec^{-1}x$ is: (A) $\left[-\frac{\pi}{2},\frac{\pi}{2}\right]$ (B) $(0,\pi)$ (C) $[0,\pi]-\left\{\frac{\pi}{2}\right\}$ (D) $\left[-\frac{\pi}{2},\frac{\pi}{2}\right]-\{0\}$	1
5.	The principal value of $\sec^{-1}(\sqrt{2})+2\csc^{-1}(-\sqrt{2})$ is: (A) $-\frac{\pi}{2}$ (B) $-\frac{\pi}{4}$ (C) $\frac{\pi}{4}$ (D) $\frac{\pi}{2}$	1

Q No.	Question	Marks
6.	**Assertion (A):** The domain of $y = \cos^{-1}x$ is $[-1,1]$. **Reason (R):** The range of the principal value branch of $y = \cos^{-1}x$ is $[0,\pi] - \left\{\frac{\pi}{2}\right\}$. (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A). (B) Both Assertion (A) and Reason (R) are true, but Reason (R) is **not** the correct explanation of the Assertion (A). (C) Assertion (A) is true, but Reason (R) is false. (D) Assertion (A) is false, but Reason (R) is true.	1
7.	Find the value of $\sin^{-1}\left(\sin\frac{3\pi}{5}\right)$.	2
	LEVEL 2 **Q 8 – Q 14**	**16**
8.	Evaluate: $\tan^{-1}\left(-\frac{1}{\sqrt{3}}\right) + \cot^{-1}\left(\frac{1}{\sqrt{3}}\right) + \tan^{-1}\left(\sin\left(-\frac{\pi}{2}\right)\right) + \tan^{-1}\left(\tan\frac{2\pi}{3}\right)$.	2
9.	Find the domain of $f(x) = \sin^{-1}(x^2 - 4)$.	2
10.	Find the domain of $f(x) = \sin^{-1}x + \cos^{-1}(x - 1)$.	2
11.	Find the value of $\cos^{-1}\left(\cos\frac{13\pi}{6}\right) + \tan^{-1}\left(\tan\frac{7\pi}{6}\right)$.	2
12.	Evaluate: $\sin\left[\frac{\pi}{3} - \sin^{-1}\left(-\frac{1}{2}\right)\right]$.	2
13.	Find the value of $\tan^{-1}\left[2\cos\left(2\sin^{-1}\frac{1}{2}\right)\right]$.	3
14.	Evaluate: $\sec^2(\tan^{-1}2) + \csc^2(\cot^{-1}3)$.	3
	LEVEL 3 **Q 15 – Q 20**	**19**
15.	Draw the graph of $y = \cos^{-1}x$ for its principal value branch. Using the graph, write its domain and range, and state whether the function is increasing or decreasing.	3
16.	Find the value of $\sin^{-1}\left(\sin\frac{4\pi}{5}\right) + \cos^{-1}\left(\cos\frac{4\pi}{3}\right) + \tan^{-1}\left(\tan\frac{3\pi}{4}\right)$.	3
17.	Find the domain and range of $f(x) = 2\sin^{-1}(1 - x)$.	3
18.	Write $\sin^{-1}\left(\frac{x}{\sqrt{1+x^2}}\right)$ in the simplest form.	3
19.	Write $\tan^{-1}\sqrt{\frac{1-\cos x}{1+\cos x}}$, $0 < x < \pi$, in the simplest form.	3
20.	**Case Study:** A lighthouse is 20 m tall. A boat is at a horizontal distance of x m from the foot of the lighthouse. The angle of elevation θ of the top of the lighthouse from the boat is given by $\theta = \tan^{-1}\left(\frac{20}{x}\right)$. **(i)** Find θ when $x = 20\sqrt{3}$ m. **[1]** **(ii)** Write the domain and range of the principal value branch of $\tan^{-1}x$. **[1]** **(iii)** Find the distance x of the boat when $\theta = \frac{\pi}{3}$. Also find the principal value of $\tan^{-1}(-\sqrt{3})$. **[2]**	4

***** End of Worksheet *****

ANSWER KEY

Chapter 2: Inverse Trigonometric Functions

Q No.	Answer / Key Steps
	LEVEL 1
1.	(A) $-\frac{\pi}{6}$
2.	(B) $\cos\frac{7\pi}{6} = -\frac{\sqrt{3}}{2}$, so value $= \frac{5\pi}{6}$.
3.	(B) $\left[-\frac{1}{2}, \frac{1}{2}\right]$
4.	(C) $[0,\pi] - \left\{\frac{\pi}{2}\right\}$
5.	(B) $\frac{\pi}{4} + 2\left(-\frac{\pi}{4}\right) = -\frac{\pi}{4}$
6.	(C) A is true; R is false (the range is $[0,\pi]$).
7.	$\sin\frac{3\pi}{5} = \sin\frac{2\pi}{5}$ and $\frac{2\pi}{5} \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$, so value $= \frac{2\pi}{5}$.
	LEVEL 2
8.	$-\frac{\pi}{6} + \frac{\pi}{3} - \frac{\pi}{4} - \frac{\pi}{3} = -\frac{5\pi}{12}$
9.	$-1 \le x^2 - 4 \le 1 \Rightarrow 3 \le x^2 \le 5$; domain $\left[-\sqrt{5}, -\sqrt{3}\right] \cup \left[\sqrt{3}, \sqrt{5}\right]$.
10.	$[-1,1] \cap [0,2] = [0,1]$
11.	$\frac{\pi}{6} + \frac{\pi}{6} = \frac{\pi}{3}$
12.	$\sin\left(\frac{\pi}{3} + \frac{\pi}{6}\right) = \sin\frac{\pi}{2} = 1$
13.	$\sin^{-1}\frac{1}{2} = \frac{\pi}{6}$; $\cos\frac{\pi}{3} = \frac{1}{2}$; $\tan^{-1}1 = \frac{\pi}{4}$.
14.	$(1 + 2^2) + (1 + 3^2) = 15$
	LEVEL 3
15.	Graph through $(-1,\pi)$, $\left(0,\frac{\pi}{2}\right)$, $(1,0)$. Domain $[-1,1]$, range $[0,\pi]$; decreasing function.
16.	$\frac{\pi}{5} + \frac{2\pi}{3} - \frac{\pi}{4} = \frac{37\pi}{60}$
17.	$-1 \le 1 - x \le 1 \Rightarrow$ domain $[0,2]$; range $[-\pi,\pi]$.
18.	Put $x = \tan\theta$, $\theta \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$: $\sin^{-1}(\sin\theta) = \theta = \tan^{-1}x$.
19.	$\sqrt{\frac{1-\cos x}{1+\cos x}} = \tan\frac{x}{2}$ with $\frac{x}{2} \in \left(0,\frac{\pi}{2}\right)$, so the value is $\frac{x}{2}$.
20.	(i) $\tan^{-1}\frac{1}{\sqrt{3}} = \frac{\pi}{6}$. (ii) Domain $\mathbb{R}$, range $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$. (iii) $\frac{20}{x} = \sqrt{3} \Rightarrow x = \frac{20}{\sqrt{3}}$ m; $\tan^{-1}\left(-\sqrt{3}\right) = -\frac{\pi}{3}$.